

1.

(a) $P(X \geq 2)$

$$P(X \geq 2) = 1 - P(X = 1) = 1 - \frac{1}{13} = \frac{12}{13}.$$

(b) $P(X \leq 10)$

$$P(X \leq 10) = 1 - P(X = 11) = \frac{10}{13}.$$

(c) $P(X \leq Y)$

- $X \leq Y$ is false for red cards, true for black.
- $P(X \leq Y) = 26/52 = 1/2$.

(d) **Distribution of $Y - X$**

$$P(Y - X = -1) = \frac{1}{2}, \quad P(Y - X = 1) = \frac{1}{2}.$$

(e) $P(Y \leq 12)$

- $Y \leq 12$ always holds (no cases where $Y > 12$).

$$P(Y \leq 12) = 1.$$

2.

(a) **Probability Distribution of X**

Using probability rules:

$$P(X = 1) = P(H) = \frac{1}{2}$$

$$P(X = 2) = P(T) \cdot P(\text{die shows 2 or 3}) = \frac{1}{2} \times \frac{2}{6} = \frac{1}{6}$$

$$P(X = 3) = P(T) \cdot P(\text{die shows 1, 4, 5, or 6}) = \frac{1}{2} \times \frac{4}{6} = \frac{2}{6} = \frac{1}{3}$$

Thus, the probability distribution is:

X	1	2	3
$P(X)$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$

(b) **Cumulative Distribution Function (CDF)**

The cumulative distribution function $F(x) = P(X \leq x)$:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{2}{3}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

This defines the probability accumulation for each x .

3.

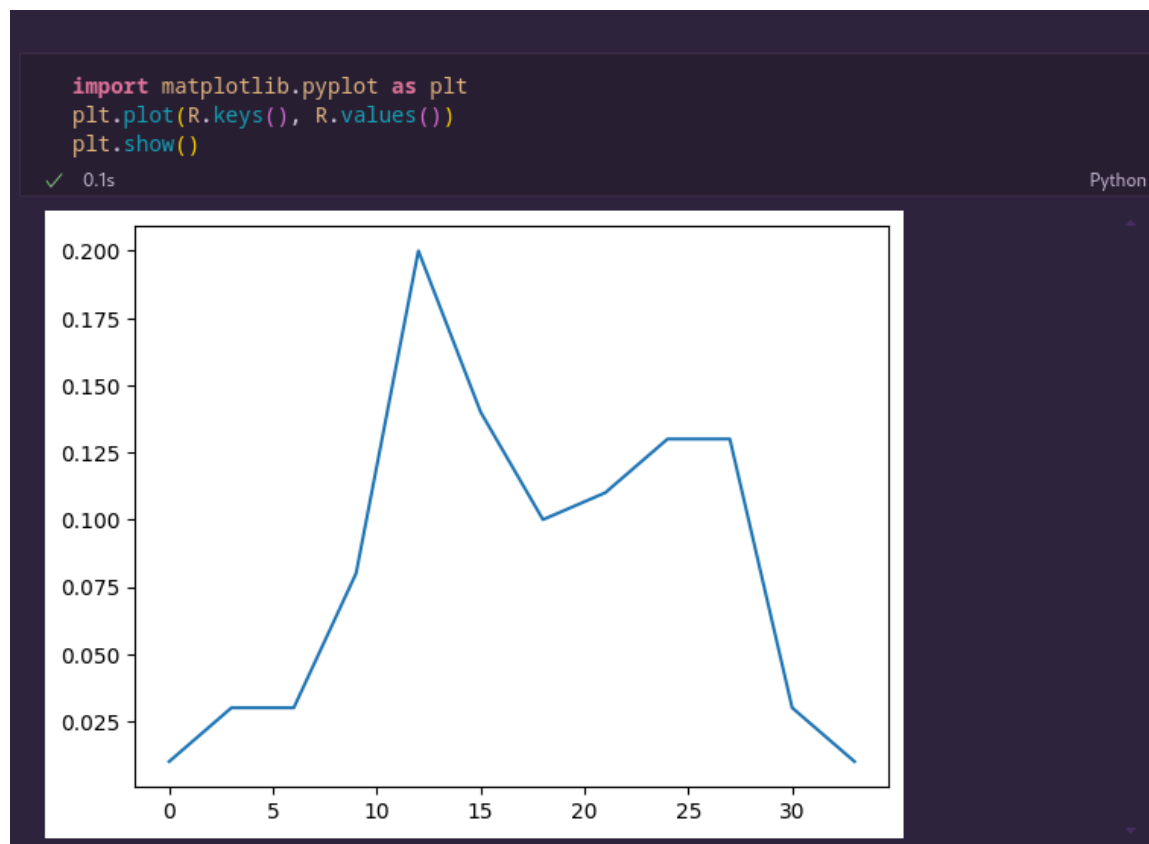
(a) The revenue for the ferry is $R = 3X + 9Y$. Find the possible values of R and the associated probabilities.

```
R: dict = {}
for x in range(6):
    for y in range(3):
        r: int = 3 * x + 9 * y
        R[r] = R.get(r, 0.0) + data[x][y]

R: dict = {k: round(v, 2) for k, v in sorted(R.items())}
R
0.0s  Open 'R' in Data Wrangler Python
```

```
{0: 0.01,
3: 0.03,
6: 0.03,
9: 0.08,
12: 0.2,
15: 0.14,
18: 0.1,
21: 0.11,
24: 0.13,
27: 0.13,
30: 0.03,
33: 0.01}
```

(b) Compute the expected revenue $E(R)$ and the expected profit for the ferry owner. What is the standard deviation of the profit?



```
expected_revenue: int = sum([revenue*probability for revenue, probability in R.items()])
print("Expected Revenue:", expected_revenue)
print("Expected Profit:", round(expected_revenue-10, 2))
```

✓ 0.0s

Python

Expected Revenue: 17.46
Expected Profit: 7.46

```
from math import sqrt
```

```
variance_revenue: int = sum((r ** 2) * p for r, p in R.items()) - expected_revenue**2
sqrt(variance_revenue)
```

✓ 0.0s

Python

7.192245824497382

4.

Sure! Let's break this down step-by-step and solve it using the **Hypergeometric distribution**.

Hypergeometric Distribution

PMF:

$$P(X = k) = \frac{\binom{D}{k} \binom{N-D}{n-k}}{\binom{N}{n}}$$

Parameters for the problem:

- $N = 25$ (total items in the box)
- $n = 3$ (sample size)
- $D = 4$ (number of defectives in the box)

(a) Probability that a box containing 4 defectives will be shipped

Using the Hypergeometric distribution:

$$P(X = 0) = \frac{\binom{4}{0} \binom{21}{3}}{\binom{25}{3}}$$

$$\binom{4}{0} = 1$$

$$\binom{21}{3} = \frac{21 \times 20 \times 19}{3 \times 2 \times 1} = 1330$$

$$\binom{25}{3} = \frac{25 \times 24 \times 23}{3 \times 2 \times 1} = 2300$$

$$P(X = 0) = \frac{1 \times 1330}{2300} \approx 0.5783$$

Therefore, the probability is 0.5783.

(b) Probability that a box containing only 1 defective will be sent back for screening

There must be **at least 1 defective** in the sample of 3 items.

Calculate the probability of $k = 0$, then subtract it from 1:

$$P(X = 0) = \frac{\binom{1}{0} \binom{24}{3}}{\binom{25}{3}}$$

$$\binom{1}{0} = 1$$

$$\binom{24}{3} = \frac{24 \times 23 \times 22}{3 \times 2 \times 1} = 2024$$

$$\binom{25}{3} = 2300 \quad (\text{from part a})$$

$$P(X = 0) = \frac{1 \times 2024}{2300} \approx 0.8800$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.8800 = 0.1200$$

Therefore, the probability is 0.1200.